

AI Vehicle PREDICTIVE MAINTENANCE

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Modern vehicle engines emit distinct sounds when operating normally or malfunctioning. AI-driven continuous monitoring detects abnormalities early, preventing costly breakdowns. This system leverages Google Teachable Machine to train an AI model to differentiate between normal and faulty engine sounds. Designed for predictive maintenance, the system offers a low-cost, efficient solution.

Teachable machine model

A machine learning model is trained to recognise normal and abnormal engine sounds. It runs in a browser, continuously monitoring the engine's sound. If an anomaly is detected, it immediately triggers an alert.

When the AI model detects an abnormal condition, it sends an HTTP request to the IndusBoard Coin. The IndusBoard Coin, acting as a web server, receives and processes the alert.

The IndusBoard Coin hosts a webpage that visually displays the system status. If a fault is detected, the page turns red and shows an alert message. Simultaneously, the IndusBoard Coin activates a blinking LED as a warning indicator.

Fig. 1 shows the webpage UI running the trained ML model. Fig. 2 shows the prototype with an IndusBoard Coin. The components required to build this system are listed in the Bill of Materials table. Fig. 3 shows the ML model being trained in Teachable Machine.

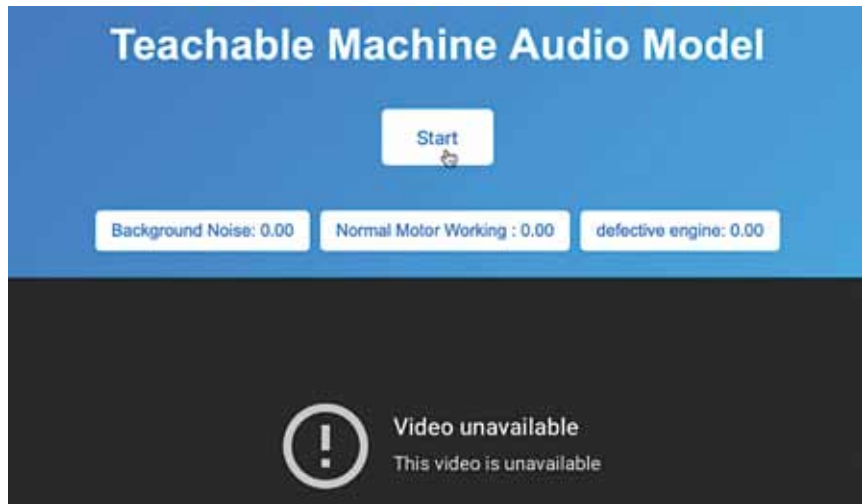


Fig. 1: Webpage UI running the trained ML model

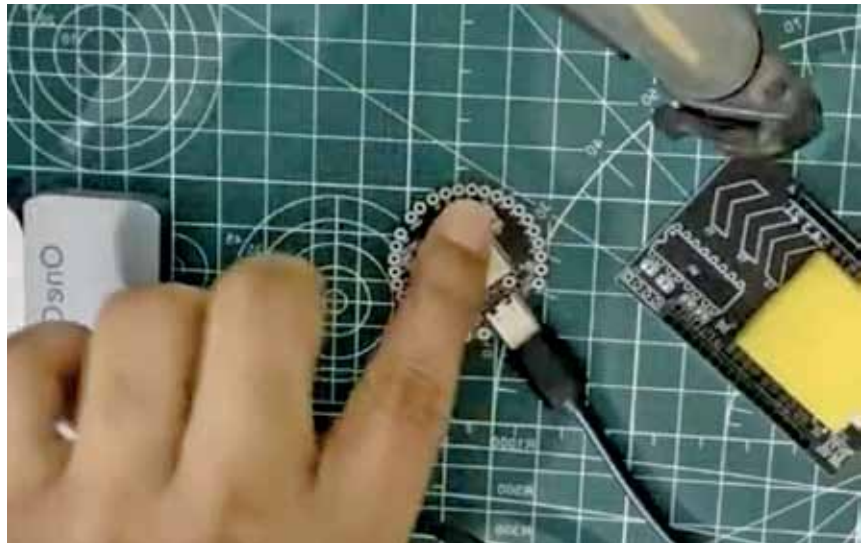


Fig. 2: Author's prototype

BILL OF MATERIALS

Component	Specification	Quantity
IndusBoard Coin	Coin V2 development board	1
Laptop	For simulation	1
Car engine	Any working vehicle engine or motor for testing	1